Supporting Information

The p53 pulse encoder counts by topological necessity: theory, evidence, and the molecular origin of the 5.5-hour clock

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This Supporting Information documents the computational output of the accompanying repository (`p53-pulse-encoder`, release v1.0.2): the analysis scripts, the archived result file for each analysis that produces one, and the provenance of third-party data. Numbers below are transcribed from the archived result files (`results/`); where a script writes only to its console, the pipeline map says so. Scripts are plain Python 3 (numpy/scipy/matplotlib). The digitised Batchelor Fig. 1G/H extraction is shipped in `data_external/batchelor2011/` and is the direct input of code52; its values and error bands are also reported in the main text.

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1 Pipeline map: scripts and outputs

Script	Computes	Archived output
code13_p53_fisher_ stratification_S7	Fisher stratification of the pulse generator, S7 slice	p53_channel_census.csv
code50_p53_cascade_ framework_selfcheck	internal consistency self-check of the cascade framework	console log
code51r1--r4_p53_ fisher_ audit	Fisher identifiability audit of the phenomenological pulse model; r4 adds noise calibration and an M=4000 central-difference control	code51r2/r3 results JSONs (code51r2_p53_Fisher_ audit_results.json, code51r3_...json), code51r4A_...json, code51r4B_Fisher_ M4000.json, three verdict figures (PNG)
code52_TrackC2_ discrimination_diagram	discrimination map between counting and amplitude encoders	code52_TrackC2_ results.json, discrimination-map PNG
code81_p53_full_ cascade_ audit	full-cascade simulation (damage → fate), information identity, amplitude-share sweep	code81_cascade_audit_ results.json, five-panel PNG
code82_p53_mechanism_ model_audit	mechanistic 6-species model; baseline reproduction, Fisher audit, mechanism tests	code82_mechanism_audit_ results.json, two four-panel PNGs
code83_why_digital_ variational_competition	variational argument for digital pulses	console/tables
code84_discovery_ cascade_model	discovery-grade cascade variant	console/tables
code85_heterogeneity_ recheck	heterogeneity robustness re-check	console/tables
code86_foci_ persistence_ trigger	DSB-foci persistence as trigger	console/tables
code87_clock_ dose_ independent_ architecture	dose-independence of the clock architecture	console/tables
code88_foci_ termination_ channel	foci-termination channel analysis	console/tables
code89_p53_molecular_ constant_decomposition	decomposition of the clock period into molecular phase constants	molecular_layer/code89_ ...json
code90_p53_full_ cascade_ consistency_ audit	14 static link closures; onset-time decomposition	molecular_layer/code90_ ...json
code91_K2_pulse_width_ decomposition	pulse-width (FWHM) molecular attribution	molecular_layer/code91_ ...json

Script	Computes	Archived output
`code92_belgacem_closedform_crossvalidation`	reproduction of the closed-form period of arXiv:2605.23722 and reconciliation	`molecular_layer/code92_...json`
`code93_tcs_logistic_correspondence`	event-time invariance under logistic time warp	`molecular_layer/code93_...json`
`cellline1_bifurcation_suite/` (scripts 1–3)	stochastic population simulation, bifurcation analysis, discrimination diagram (cell-line 1 suite cited in Methods)	figures
Manuscript figures (all in `manuscript/`)		
`make_fig1.py`	Fig. 1 encoder topology	`fig1_topology.png`
`make_fig2.py`	Fig. 2 channel census	`fig2_census.png`
`make_fig3.py`	Fig. 3 molecular decomposition (figure layer of code89; reads `code89_...json`)	`fig3_molecular.png`
`make_fig4.py`	Fig. 4 closed-form cross-validation (figure layer of code92; reads `code92_...json`)	`fig4_closedform.png`

Run metadata: code81/82 seed `20260923`; code89–93 seed `20260925`; Fisher audits use relative step $h \in \{2\%, 5\%\}$ with a central-difference control at M=4000 trajectories.

2 Fisher identifiability audit (code51r1–r4)

Operating point θ_0 of the phenomenological model (Table 2); observables: t_1, T, W, amplitude, and the pulse-count distribution.

Table 2: Operating point used in all Fisher audits.

a	b	ϵ	s	τ_r (min)	I_0	σ
0.7	0.8	0.08	20.0	20.0	0.6	0.06

Table 3: Condition numbers of the Fisher matrix across audits.

Audit	Setting	$\kappa(F)$
code51r1	h=2% / 5%	∞ (exact null direction, k_{obs})
code51r2	h=2%	7.52×10^{17}
code51r2	h=5%	1.38×10^{4}
code51r3	h=2%	4.87×10^{17}
code51r3	h=5%	4.37×10^{6}
code51r4B	M=4000, central, h=2%	2.13×10^{18}

Supporting quantities: mean pulse count $\bar{N} = 9.04$ (r1/r3), 8.33 (r2), 9.01 (r4B); $\mathrm{var}(t_1) = 1.29 \times 10^{-4}$, $\mathrm{var}(T) = 1.09 \times 10^{-3}$, $\mathrm{var}(t_1/T) = 0.119$. Noise calibration (code51r4A): adopted $\sigma^\star = 0.25$, giving

$\mathrm{CV}_T = 0.111$, IPI CV $= 0.316$, valid fraction 1.0, $\bar{N} = 10.39$. Channel capacity across 9 doses: $I(D;N) = 3.07$ bits vs $H(D) = 3.17$ bits (code51r3).

3 Counting-vs-amplitude discrimination (code52)

Slope estimators on the Track-C2 discrimination diagram (Table 4).

Table 4: Discrimination slopes (mean ± s.d., 95% CI).

Arm	mean	s.d.	95% CI
α_{NCS}	-0.0008	0.113	$(-0.225,\ 0.222)$
α_{UV}	0.981	0.114	$(0.777,\ 1.224)$

The τ_r inversion point sits at 7.7 h.

4 Full-cascade audit (code81)

Simulation: 7 doses × 4000 cells/dose. The 20 frozen constants are listed in Table 5.

Table 5: Frozen constants of the code81 cascade (verbatim from the archive).

Constant	Value	Constant	Value
$K0_{\mathrm{DSB}}$	35.0	Q_{21} / τ_{21}	100.0 / 10.0
C_{CLUSTER}	50.0	Q_{PU} / τ_{PU}	100.0 / 4.0
N_{MAX}	12	PUMA n_{thr}	3
T_{PERIOD} (h)	5.5	θ_{PUMA}	800.0
μ_D/T	1.0	$R0_{\mathrm{FATE}}$	0.15
$\mathrm{JIT}_{\mathrm{IPI}}$	0.085	θ_{P21}	1500.0
A_0	1.0	B_{GAIN}	0.3
A_{CV}	0.002	η_{OBS}	0.02
W (h)	48.0	α_E	7.0

Table 6: Fate table (fraction of 4000 cells) vs dose.

Dose (Gy)	survival	arrest	apoptosis
0.1	1.0000	0.0000	0.0000
0.25	1.0000	0.0000	0.0000
0.5	1.0000	0.0000	0.0000
1.0	0.9948	0.0053	0.0000
2.5	0.0047	0.9952	0.0000
5.0	0.0015	0.9688	0.0297
10.0	0.0020	0.0088	0.9892

Information identity: $H(D) = 2.8074$ bits decomposes into the five layer contributions with closure residual 0.0; the weakest layer is L1 (sensing), losing 0.872 bits. Amplitude share of dose information

at kinetic-noise levels $\sigma_k = 0.0/0.3/0.6$: 34.5%/14.8%/9.9% of the counting information ($I_{\text{count}} = 1.631$ bits throughout).

5 Mechanistic-model audit (code82)

Six-species model (ATM*, p53, mdm2 mRNA, Mdm2, wip1 mRNA, Wip1); 23 parameters, audited in the 2025 and 2017 parameter sets (Table 7).

Table 7: Parameter sets (code82 archive).

Set	A	P	C	g	d_{AM}	T_m	T_M	T_w	T_W	d_A	d_P	d_m
2025	30.5	22	1.4	2.5	20	1.2	4	1.2	1	0.16	0.1	1
2017	30.5	22	1.4	2.5	20	1.0	4	1.0	1	0.16	0.1	1

	d_M	d_w	d_W	k_A	k_{WA}	k_{MP}	k_{Pm}	k_{Pw}	R	$S_{\max}$	γ
2025	2	1.3	2.3	0.5	0.14	0.15	1	1	2	0.2	9
2017	2	1.3	2.3	0.5	0.14	0.15	1	1	2	0.2	9

Ground state (both sets): ATM* 0.0353, p53 0.2180, mdm2 0.2147, Mdm2 0.3174, wip1 0.1652, Wip1 0.0718.

Table 8: R1 baseline reproduction: pulse count, period, amplitude.

Set	Damage input	pulses	period (h)	amplitude
2025	log	17	5.484	2.783
2025	saturating	17	5.753	2.788
2017	log	14	6.831	3.408
2017	saturating	14	6.681	3.407

Damage-signal integral $S = 0.4988$ (log) / 0.1982 (saturating). The stiffest Fisher axis is the Wip1-production axis ($T_w\,0.27 + T_W\,0.26 + P\,0.11 + d_W\,0.10$).

6 Molecular decomposition of the 5.5 h clock (code89)

Angular frequency $\omega = 1.1424$ rad/h.

Table 9: Phase-constant decomposition vs measurement.

Quantity	Predicted	Target
T_1: K3 mdm2 lag (h)	1.70	2.0 ± 0.5
T_2: K4 wip1 delay (h)	1.13	1.25
T_3: K1 period (h)	5.95	5.48

Per-species phase shares of the period: p53 30.4%, ATM* 24.3%, mdm2 mRNA 18.2%, wip1 mRNA 15.3%, Wip1 9.6%, Mdm2 2.2% (sum of phases = 4.474 rad). Molecular closure register: T_W (translation, universal ∼4 aa/s) and d_W (Wip1 degradation, 0.7–2.3/h across sources) are measurable constants; T_w and P are registered gaps.

7 Full-cascade consistency (code90)

14 static links audited: **7 closed, 3 tension, 4 gap.** Onset decomposition at the step signal ($S = 0.4988$, basal 0.0401): sensing contributes only 0.83 h in total (γH2AX segment 0.50 h, ATM half-rise 0.19 h, p53 threshold crossing 0.14 h); the first p53 peak arrives at 3.21 h, i.e. the bulk of the measured t_1 (MCF7 band 2–3 h) is the accumulation ramp (2.38 h), not sensing delay.

8 Pulse-width decomposition (code91)

At DSB = 100: measured FWHM $W = 2.779$ h decomposes as p53 half-rise $\rightarrow$ ATM* extinction (0.79 h) + collapse tail (1.99 h). The plateau segment (1.54 h) is set by Wip1 accumulation; the collapse tail by Mdm2 recovery kinetics (d_M/d_{AM}) — the dominant molecular attribution of width is Mdm2 recovery, not the Wip1 threshold crossing itself. Dose scan: width CV = 0.055.

9 Closed-form cross-validation (code92)

Reproduction of the closed form of Belgacem (arXiv:2605.23722, Section 9): $AB = 1.7204$ inverted from the manuscript constants vs 1.72 published; $\tau_c = 1.000$ h; closed-form period $T = 5.64$ h vs 5.6 h published and 5.5 h measured. Reconciliation row: measured 5.5 h; Belgacem closed form 5.64 h; Monke simulation 5.48 h; Monke Jacobian 5.95 h; our τ_2 closed-form evaluation 8.40 h (registered as a tension, see the main text).

10 TCS–logistic correspondence (code93)

Event-time invariance under the logistic time warp: 7 pulses before and after warping, maximal event-time shift 0.0 h, period unchanged at 5.4802 h, while the amplitude CV inflates by +42.3% (0.308 $\rightarrow$ 0.438).

11 Third-party data provenance

No large third-party archive is re-hosted. Extracted tables in `results/` trace to: Batchelor et al., *Mol. Syst. Biol.* 7, 488 (2011); Stewart-Ornstein & Lahav, *Sci. Signal.* 10, eaah6671 (2017) (PMC5504473); Geva-Zatorsky et al., *Mol. Syst. Biol.* 2, 33 (2006); Loewer et al., *Cell* 142, 89–100 (2010) (PMC3003696); I. Belgacem, arXiv:2605.23722 (2026). Molecular-layer constants (delays, half-lives) trace to Wang et al., *IET Syst. Biol.* 13, 180 (2019); Monk, *Curr. Biol.* 13, 1409 (2003); Batchelor et al., *Mol. Cell* 30, 277 (2008); Mönke et al., *Sci. Rep.* 7, 46571 (2017). Full access notes in `data_external/PROVENANCE.md`.